

Exercises for Sect. 2.4

Exercise 2.4.1. Give a different proof that the circle rotation $R_\alpha : \mathbb{T} \rightarrow \mathbb{T}$ is ergodic if α is irrational, using Lebesgue's density theorem (Theorem A.24) as follows. Suppose if possible that A and B are measurable invariant sets with $0 < m_{\mathbb{T}}(A), m_{\mathbb{T}}(B) < 1$ and $A \cap B = \emptyset$, and use the fact that the orbit of a point of density for A is dense to show that $A \cap B$ must be non-empty.

Exercise 2.4.2. Prove that an ergodic toral automorphism is not measurably isomorphic to an ergodic circle rotation.

Exercise 2.4.3. Extend Proposition 2.16 as follows. If X is a compact abelian group, prove that the group rotation $R_g(x) = gx$ is ergodic with respect to Haar measure if and only if the subgroup $\{g^n \mid n \in \mathbb{Z}\}$ generated by g is dense in X .

Exercise 2.4.4. In the notation of Corollary 2.20, prove that A is injective if and only if $|\det(A)| = 1$, and in general that $A : \mathbb{T}^d \rightarrow \mathbb{T}^d$ is $|\det(A)|$ -to-one if $\det(A) \neq 0$. Prove Corollary 2.20 using Theorem 2.19 and the explicit description of characters on the torus from (C.3) on p. 436.

2.5 The Mean Ergodic Theorem

Ergodic theorems at their simplest express a relationship between averages taken along the orbit of a point under iteration of a measure-preserving map (in the physical origins of the subject, this represents an average over *time*) and averages taken over the measure space with respect to some invariant measure (an average over *space*). The averages taken are of *observables* in the physical sense, represented in our setting by measurable functions. Much of this way of viewing dynamical systems goes back to the seminal work of von Neumann [268].

We have already seen that ergodicity is a spectral property; the first and simplest ergodic theorem only uses properties of the operator U_T associated to a measure-preserving transformation T . Theorem 2.21 is due to von Neumann [267] and predates⁽¹⁸⁾ the pointwise ergodic theorem (Theorem 2.30) of Birkhoff, despite the dates of the published versions.

Write $\xrightarrow{L_\mu^p}$ for convergence in the L_μ^p norm.

Theorem 2.21 (Mean Ergodic Theorem). *Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system, and let P_T denote the orthogonal projection onto the closed subspace*

$$I = \{g \in L_\mu^2 \mid U_T g = g\} \subseteq L_\mu^2.$$

Then for any $f \in L_\mu^2$,

$$\frac{1}{N} \sum_{n=0}^{N-1} U_T^n f \xrightarrow{L_\mu^2} P_T f.$$

PROOF. Let $B = \{U_T g - g \mid g \in L_\mu^2\}$. We claim that $B^\perp = I$. If

$$U_T f = f,$$

then

$$\langle f, U_T g - g \rangle = \langle U_T f, U_T g \rangle - \langle f, g \rangle = 0,$$

so $f \in B^\perp$. If

$$f \in B^\perp$$

then

$$\langle U_T g, f \rangle = \langle g, f \rangle$$

for all $g \in L_\mu^2$, so

$$U_T^* f = f. \quad (2.9)$$

Thus

$$\begin{aligned} \|U_T f - f\|_2 &= \langle U_T f - f, U_T f - f \rangle \\ &= \|U_T f\|_2^2 - \langle f, U_T f \rangle - \langle U_T f, f \rangle + \|f\|_2^2 \\ &= 2\|f\|_2^2 - \langle U_T^* f, f \rangle - \langle f, U_T^* f \rangle \\ &= 0 \quad \text{by (2.9),} \end{aligned}$$

so $f = U_T f$.

It follows that $L_\mu^2 = I \oplus \overline{B}$, so any $f \in L_\mu^2$ decomposes as

$$f = P_T f + h, \quad (2.10)$$

with $h \in \overline{B}$. We claim that

$$\frac{1}{N} \sum_{n=0}^{N-1} U_T^n h \xrightarrow{L_\mu^2} 0.$$

This is clear for $h = U_T g - g \in B$, since

$$\begin{aligned} \left\| \frac{1}{N} \sum_{n=0}^{N-1} U_T^n (U_T g - g) \right\|_2 &= \left\| \frac{1}{N} ((U_T g - g) + (U_T^2 g - U_T g) + \cdots \right. \\ &\quad \left. + (U_T^N g - U_T^{N-1} g)) \right\|_2 \\ &= \frac{1}{N} \|U_T^N g - g\|_2 \longrightarrow 0 \end{aligned} \quad (2.11)$$

as $N \rightarrow \infty$. All we know is that $h \in \overline{B}$, so let (g_i) be a sequence in L_μ^2 with the property that $h_i = U_T g_i - g_i \rightarrow h$ as $i \rightarrow \infty$. Then for any $i \geq 1$,

$$\left\| \frac{1}{N} \sum_{n=0}^{N-1} U_T^n h \right\|_2 \leq \left\| \frac{1}{N} \sum_{n=0}^{N-1} U_T^n (h - h_i) \right\|_2 + \left\| \frac{1}{N} \sum_{n=0}^{N-1} U_T^n h_i \right\|_2. \quad (2.12)$$

Fix $\varepsilon > 0$ and choose, by the convergence (2.11), quantities i and N so large that

$$\|h - h_i\|_2 < \varepsilon$$

and

$$\left\| \frac{1}{N} \sum_{n=0}^{N-1} U_T^n h_i \right\|_2 < \varepsilon.$$

Using these estimates in the inequality (2.12) gives

$$\left\| \frac{1}{N} \sum_{n=0}^{N-1} U_T^n h \right\|_2 \leq 2\varepsilon$$

so

$$\frac{1}{N} \sum_{n=0}^{N-1} U_T^n h \xrightarrow{L_\mu^2} 0$$

as $N \rightarrow \infty$, for any $h \in \overline{B}$. The theorem follows by (2.10). $\square$

The quantity studied in Theorem 2.21 is an *ergodic average*, and it will be convenient to fix some notation for these. For a fixed measure-preserving system $(X, \mathcal{B}, \mu, T)$ and a function $f : X \rightarrow \mathbb{C}$ the N th ergodic average of f is defined to be

$$A_N = A_N^f = A_N(f) = \frac{1}{N} \sum_{n=0}^{N-1} f \circ T^n.$$

It is important to understand that this will be interpreted in several quite different ways.

- In Theorem 2.21 the function f is an element of the Hilbert space L_μ^2 (that is, an equivalence class of measurable functions) and A_N^f is thought of as an element of L_μ^2 .
- In Corollary 2.22 we will want to think of f as an element of L_μ^1 , but evaluate the ergodic average A_N^f at points, sometimes writing $A_N^f(x)$. Of course in this setting any statement can only be made almost everywhere with respect to μ , since f (and hence A_N^f) is only an equivalence class of functions, with two point functions identified if they agree almost everywhere.

- At times it will be useful to think of f as an element of $\mathcal{L}_\mu^p$ (that is, as a function rather than an equivalence class of functions) in which case A_N^f is defined everywhere. Also, if f is continuous, we will later ask whether the convergence of $A_N^f(x)$ could be uniform across $x \in X$.

Corollary 2.22. ⁽¹⁹⁾ *Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system. Then for any function $f \in L_\mu^1$ the ergodic averages A_N^f converge in L_μ^1 to a T -invariant function $f' \in L_\mu^1$.*

PROOF. By the mean ergodic theorem (Theorem 2.21) we know that for any $g \in L_\mu^\infty \subseteq L_\mu^2$, the ergodic averages A_N^g converge in L_μ^2 to some $g' \in L_\mu^2$. We claim that $g' \in L_\mu^\infty$. Indeed, $\|A_N^g\|_\infty \leq \|g\|_\infty$ and so

$$|\langle A_N^g, \chi_B \rangle| \leq \|g\|_\infty \mu(B)$$

for any $B \in \mathcal{B}$. Since $A_N^g \rightarrow g'$ in L_μ^2 , this implies that

$$|\langle g', \chi_B \rangle| \leq \|g\|_\infty \mu(B)$$

for $B \in \mathcal{B}$, so $\|g'\|_\infty \leq \|g\|_\infty$ as required.

Moreover, $\|\cdot\|_1 \leq \|\cdot\|_2$, so we deduce that

$$A_N^g \xrightarrow{L_\mu^1} g' \in L_\mu^\infty.$$

Thus the corollary holds for the dense set of functions $L_\mu^\infty \subseteq L_\mu^1$.

Let $f \in L_\mu^1$ and fix $\varepsilon > 0$; choose $g \in L_\mu^\infty$ with $\|f - g\|_1 < \varepsilon$. By averaging,

$$\left\| \frac{1}{N} \sum_{n=0}^{N-1} f \circ T^n - \frac{1}{N} \sum_{n=0}^{N-1} g \circ T^n \right\|_1 < \varepsilon,$$

and by the previous paragraph there exists g' and N_0 with

$$\left\| \frac{1}{N} \sum_{n=0}^{N-1} g \circ T^n - g' \right\|_1 < \varepsilon$$

for $N \geq N_0$. Combining these gives

$$\left\| \frac{1}{N} \sum_{n=0}^{N-1} f \circ T^n - \frac{1}{N'} \sum_{n=0}^{N'-1} f \circ T^n \right\|_1 < 4\varepsilon$$

whenever $N, N' \geq N_0$. In other words, the ergodic averages form a Cauchy sequence in L_μ^1 , and so they have a limit $f' \in L_\mu^1$ by the Riesz–Fischer theorem (Theorem A.23). Since

$$\left\| \left(\frac{1}{N} \sum_{n=0}^{N-1} f \circ T^n \right) \circ T - \frac{1}{N} \sum_{n=0}^{N-1} f \circ T^n \right\|_1 < \frac{2}{N} \|f\|_1$$

for all $N \geq 1$, the limit function f' must be T -invariant. $\square$

Exercises for Sect. 2.5

Exercise 2.5.1. Show that a measure-preserving system $(X, \mathcal{B}, \mu, T)$ is ergodic if and only if, for any $f, g \in L^2_\mu$,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} \langle U_T^n f, g \rangle = \langle f, 1 \rangle \cdot \langle 1, g \rangle.$$

Exercise 2.5.2. Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system. For any function f in L^p_μ , $1 \leq p < \infty$, prove that

$$\frac{1}{n} \sum_{i=0}^{n-1} f(T^i x) \xrightarrow{L^p_\mu} f^*,$$

with $f^* \in L^p_\mu$ a T -invariant function.

Exercise 2.5.3. Show that a measure-preserving system $(X, \mathcal{B}, \mu, T)$ is ergodic if and only if $A_N(f) \rightarrow \int f d\mu$ as $N \rightarrow \infty$ for all f in a dense subset of L^1_μ .

Exercise 2.5.4. Extend Theorem 2.21 to a uniform mean ergodic theorem as follows. Under the assumptions and with the notation of Theorem 2.21, show that

$$\lim_{N-M \rightarrow \infty} \frac{1}{N-M} \sum_{n=M}^{N-1} U_T^n f \rightarrow P_T f.$$

Exercise 2.5.5. Apply Exercise 2.5.4 to strengthen Poincaré recurrence (Theorem 2.11) as follows. For any set B of positive measure in a measure-preserving system $(X, \mathcal{B}, \mu, T)$,

$$E = \{n \in \mathbb{N} \mid \mu(B \cap T^{-n}B) > 0\}$$

is syndetic: that is, there are finitely many integers $k_1, \dots, k_s$ with the property that $\mathbb{N} \subseteq \bigcup_{i=1}^s E - k_i$.

Exercise 2.5.6. Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system. We say that T is *totally ergodic* if T^n is ergodic for all $n \geq 1$. Given $K \geq 1$ define a space $X^{(K)} = X \times \{1, \dots, K\}$ with measure $\mu^{(K)} = \mu \times \nu$ defined on